

5(a)(i)	(they are) perpendicular	B1
5(a)(ii)	(they are) parallel	B1
5(b)(i)	$\lambda = v / f$	C1
	$= 340 / 1700$ $= 0.20 \text{ m}$	A1
5(b)(ii)	$L = \frac{3}{4} \times \lambda = \frac{3}{4} \times 0.20$ $= 0.15 \text{ m}$	A1
5(b)(iii)	$\lambda = 4 \times 0.15$ or 0.20×3 $= 0.60 \text{ m}$	A1
5(c)(i)	$(I =) 8.5 \times \cos^2 35^\circ = 5.7 \text{ (W m}^{-2}\text{)}$	A1
5(c)(ii)	$5.2 = 5.7 \cos^2 \theta$ $(\theta = 17^\circ)$	C1
	$\alpha = 35^\circ + 17^\circ$ $= 52^\circ$	A1